

Biographical Sketch

Gary Leo Welz

Principal Investigator, NSF CISE CORE (IIS) Proposal

Professional Preparation

Bedford College, University of London, London, England - M.Sc. Mathematics, 1977

Bedford College, University of London, London, England - B.A. Philosophy & Mathematics, 1976

Appointments

AI Software Developer / Independent Researcher (2023-Present) - Retired member, CUNY Graduate Center New Media Lab - Developer of CopernicusAI Knowledge Engine and related AI knowledge management tools

Mathematics Instructor (Adjunct Lecturer) (2006-2020) - Borough of Manhattan Community College, New York, NY - Taught Developmental Mathematics, Quantitative Reasoning, Introduction to Statistics

Mathematics Instructor (1988-1995, 2002-2018) - John Jay College of Criminal Justice, CUNY, New York, NY - Adjunct Lecturer and Substitute Instructor - Taught Developmental Mathematics, Finite Mathematics, Pre-Calculus, College Algebra, Calculus, Survey of Concepts of Operations Research, Discrete Mathematics

Senior Business Consultant, Media Services Group (2000-2001) - EMC Corporation, New York, NY - High-level business consulting to media companies in digital media production, broadband content distribution, and media asset management

Electronic Media Journalist (1994-2002) - Freelance Feature Article Writer and Columnist - Wrote feature articles for Internet World, Web Week, ACM Interactions, The X Journal, The X Advisor

Products Most Closely Related to Proposed Project

Publications

A Vision for AI-Powered Knowledge Engines: A Framework for Systematic Knowledge Discovery and Integration (2025) - *Nature Machine Intelligence* (under review; Perspective) - Preprint: <https://doi.org/10.5281/zenodo.18463304> - Proposes “Knowledge Engines” as a framework for systems that systematically transform information into knowledge; nine-capability taxonomy; presents CopernicusAI as working implementation

Software and Digital Platforms (2023-Present)

CopernicusAI Knowledge Engine (2023-Present) - Main Platform: <https://www.copernicusai.fyi> - Hugging Face Space: <https://huggingface.co/spaces/garywelz/copernicusai> - AI-powered research briefing and podcast generation system - Operational platform with generated podcast episodes across multiple scientific disciplines - Integration with multiple academic databases providing access to research papers - Multi-voice AI podcast

generation with RSS feed distribution - Evidence-based content requiring minimum 3 research sources per briefing

Programming Framework (2023-Present) - Hugging Face Space: https://huggingface.co/spaces/garywelz/programming_framework
- Zenodo: <https://doi.org/10.5281/zenodo.18463442> - Universal method for process analysis using LLM-powered extraction and Mermaid visualization - Domain-agnostic methodology demonstrated across multiple scientific disciplines - Converts textual process descriptions into structured, computable diagrammatic representations

Genome Logic Modeling Project (GLMP) (2023-Present) - Hugging Face Space: https://huggingface.co/spaces/garywelz/genome_logic_modeling
- First specialized application of Programming Framework to biological process visualization - Biological processes mapped as interactive flowcharts - JSON-based structured data in Google Cloud Storage - Demonstrates feasibility of process visualization as computable research artifacts

Research Paper Metadata Database (2024-Present) - Hugging Face Space: https://huggingface.co/spaces/garywelz/research_paper_metadata
- GitHub Repository: <https://github.com/garywelz/copernicusai-research-metadata> - Centralized metadata repository with AI-powered preprocessing - Citation network analysis and structured JSON storage - Foundation for unified multi-modal metadata representation

Science Video Database (2024-Present) - Hugging Face Space: <https://huggingface.co/spaces/garywelz/sciencevideodb>
- Live Demo: <https://scienceviddb-web-204731194849.us-central1.run.app/> - Curated searchable database of scientific video content - Transcript-based search across multiple scientific disciplines - Demonstrates multi-modal content integration for research

Earlier Scientific Communication Projects

Science Television Company (1990s) - Founder/Producer of multimedia scientific lectures - Distributed by American Mathematical Society - Current archive: <https://www.youtube.com/@scimedtv238>

The Art of Renaissance Science: Galileo and Perspective (1995) - Web multimedia presentation with CUNY Graduate Center historian Joseph W. Dauben - Hosted at Los Alamos National Lab and Padua Observatory

Other Significant Products

Publications

Wave Mathematics: Trigonometric Functions (2006) - Visionlearning.com mathematics module - <https://www.visionlearning.com/en/library/Math-in-Science/62/Wave-Mathematics/131/> - Part of Prof. Anthony Carpi's Visionlearning project, John Jay College Science Department

Is the Genome Like a Computer Program? (1996) - The X Advisor, May 1996

New Dimensions: A Multimedia Revolution on the Net (1995) - Internet World, March 1995

Video and Television on the Internet (1995) - The X Advisor, December 1995

Phase Transition: The Online Medium Comes of Age (1995) - The X Advisor, August 1995

The Media Business on the WWW (1994) - Electronic Proceedings of the 2nd International WWW Conference, October 1994

Multimedia, Hypermedia and Television on the Internet (1994) - ACM Interactions, July 1994

Television on the Internet: Scientific Publishing in a New Medium (1992) - Proceedings of the 7th Conference of the International Federation of Science Editors, 1992

Grants and Projects

Science and Engineering Television Network Project (1991-1992) - Principal Investigator - \$60,000 grant from Alfred P. Sloan Foundation to John Jay College - Feasibility study for a television network for scientists

Rubin Museum Fellow (2013) - John Jay College Learning Community program - Developed and executed integrated Mathematics and English composition program - First-year Mathematics course integrated with writing instruction

Synergistic Activities

Teaching and Educational Technology

- Implemented web-based mathematics learning systems including Maple, Mathematica, Excel, ALEKS, MyMathLab, WebAssign, and Khan Academy
- Developed innovative approaches to mathematics education using software and online tools
- Created multimedia educational content for scientific communication

Scientific Communication and Media Production

- Pioneer in internet-based scientific publishing and multimedia content distribution
- Early work on television and video distribution over the internet (1990s)
- Producer of scientific multimedia content and educational materials

AI and Knowledge Management (Current Focus)

- Development of AI-powered tools for scientific research synthesis
- Integration of large language models with structured data and visualization
- Creation of multi-modal knowledge management systems
- Open-source contributions to scientific research tools
- Public demonstration of AI-assisted research workflows through operational prototypes

Academic Affiliations

- **CUNY Graduate Center - New Media Lab:** Affiliate member (retired faculty status)
 - **John Jay College of Criminal Justice:** Retired faculty, Mathematics and Computer Science Department
 - **Borough of Manhattan Community College:** Former adjunct lecturer
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Relevant Expertise Summary

AI/ML Systems Development: - Multi-model LLM orchestration (Google Gemini, OpenAI GPT-4, Anthropic Claude) - Natural language processing for scientific content - Knowledge graph and metadata system design - Multi-modal content integration - Evidence-grounded AI system design

Software Development: - Full-stack development (Python, TypeScript, JavaScript) - Cloud infrastructure (Google Cloud Platform, serverless architectures) - Database design (NoSQL, structured metadata) - API integration and system architecture - Production-scale system deployment

Scientific Communication: - Multimedia content production and distribution - Scientific video and podcast production - Educational technology and learning systems - Research synthesis and knowledge management - Multi-modal knowledge representation

Mathematics and Education: - Mathematics instruction at college level - Educational software implementation - Curriculum development and integration - Quantitative reasoning and statistical methods

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Format: NSF Biographical Sketch (2-page limit) **Prepared For:** NSF CISE CORE (IIS) Proposal Submission
Date: December 22, 2025